
Research Articles: Behavioral/Cognitive

The pain of sleep loss: A brain characterization in humans

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<https://doi.org/10.1523/JNEUROSCI.2408-18.2018>

Received: 17 September 2018

Revised: 6 December 2018

Accepted: 7 December 2018

Published: 28 January 2019

Author contributions: A.J.K. designed research; A.J.K. performed research; A.J.K. analyzed data; A.J.K. wrote the first draft of the paper; A.J.K., A.A.P., T.D.W., M.A.L., and M.P.W. edited the paper; A.J.K. and M.P.W. wrote the paper.

Conflict of Interest: The authors declare no competing financial interests.

We would like to thank the following research assistants: Samika Kumar, Harlene Grewel, Grace Kim, Kaycee Ching, Harshika Chowdhary, Jessica Kendall-Bar, Mandana Mostofi, Jay Gupta.

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Cite as: J. Neurosci 2019; 10.1523/JNEUROSCI.2408-18.2018

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Number of Figures: 5; **Number of Tables:** 1

17

Word Counts: Abstract: 231; Introduction: 662; Discussion: 1424

18

Competing interests: The authors declare they have no competing interests.

19

Acknowledgments: We would like to thank the following research assistants: Samika

20

Kumar, Harlene Grewel, Grace Kim, Kaycee Ching, Harshika Chowdhary, Jessica

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Kendall-Bar, Mandana Mostofi, Jay Gupta.

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Abstract

Sleep loss increases the experience of pain. However, the brain mechanisms underlying altered pain processing following sleep deprivation are unknown. Moreover, it remains unclear whether ecologically modest night-to-night changes in sleep, within an individual, confer consequential day-to-day changes in experienced pain. Here, we demonstrate that acute sleep-deprivation amplifies pain reactivity within human (male and female) primary somatosensory cortex yet blunts pain-reactivity in higher-order valuation and decision-making regions of the striatum and insula cortex. Consistent with this altered neural signature, we further show that sleep deprivation expands the temperature range for classifying a stimulus as painful, specifically through a lowering of pain thresholds. Moreover, the degree of amplified reactivity within somatosensory cortex following sleep deprivation significantly predicts this expansion of experienced pain across individuals. Finally, outside of the laboratory setting, we similarly show that even modest nightly changes in sleep quality (increases and decreases) *within* an individual determine consequential day-to-day changes in experienced pain (decreases and increases, respectively). Together, these data provide a central brain framework underlying the impact of sleep loss on pain, and furthermore, establish that the association between sleep and pain is expressed in a night-to-day, bidirectional relationship within a sample of the general population. More broadly, our findings highlight sleep as a novel therapeutic target for pain management within and outside the clinic, including circumstances where sleep is frequently short yet pain is abundant (e.g. the hospital setting).

46 **Significance Statement**

47 *Are you experiencing pain? Did you have a bad night of sleep? This study*
48 *provides underlying brain and behavioral mechanisms explaining this common co-*
49 *occurrence. We show that sleep deprivation enhances pain responsivity within the*
50 *primary sensing regions of the brain's cortex yet blunts activity in other regions that*
51 *modulate pain processing—the striatum and insula. We further establish that even*
52 *subtle night-to-night changes in sleep in a sample of the general population predict*
53 *consequential day-to-day changes in pain (bidirectionally). Considering the societal rise*
54 *in chronic pain conditions in lock-step with the decline in sleep time through the*
55 *industrial world, our data support the hypothesis that these two trends may not simply*
56 *be co-occurring but are significantly inter-related.*

57

58

59 Introduction

60 Sleep loss amplifies the experience of pain. Causal manipulations in animals
61 demonstrate that sleep deprivation enhances nociceptive behavior (Lautenbacher et al.,
62 2006). Experiments in humans establish that total (Schuh-Hofer et al., 2013), partial
63 (Faraut et al., 2015), and selective sleep deprivation (Lentz et al., 1999; Roehrs et al.,
64 2006) increase pain, including a lowering of pain thresholds (Schimpf et al., 2015).

65 Despite the robust links, the central brain mechanisms underlying the impact of
66 sleep loss on pain perception remain unknown, in any species. Beyond gross
67 manipulations of sleep within the laboratory, it is similarly unknown whether modest,
68 societally common, night-to-night changes in sleep *within* an individual confer
69 consequential day-to-day changes in experienced pain. This study sought to address
70 both of these currently unanswered questions.

71 Though no reports have explored the brain basis of sleep-loss induced
72 hyperalgesia, neuroimaging studies in non-sleep-deprived individuals have established
73 a pattern of fMRI activity in regions responsive to pain (Brown et al., 2011; Wager et al.,
74 2013) reviewed in (Tracey and Mantyh, 2007; Woo et al., 2017). These include primary
75 and secondary somatosensory cortices that map noxious stimuli (Apkarian et al., 2005;
76 Atlas et al., 2014). Activity within somatosensory cortex scales as a function of pain
77 intensity, including at pain threshold (Coghill et al., 1999; Timmermann et al., 2001;
78 Apkarian et al., 2005; Atlas et al., 2014).

79 Corticolimbic regions of the cingulate and insula are consistently activated by
80 pain and are involved in the second-order mapping of internal state changes associated
81 with pain (Craig, 2002). Subcortical regions including spinal-afferent-receiving regions of
82 the thalamus, as well as nucleus accumbens (NAcc), also show robust pain reactivity
83 (Navratilova and Porreca, 2014). The NAcc in particular plays a role in encoding
84 affective value and saliency, including that of pain (Redgrave et al., 2008; Schultz,
85 2013; Navratilova and Porreca, 2014), allowing the modulation of ongoing pain through
86 descending brain and spinal cord pathways (Navratilova and Porreca, 2014).

87 This pain-reactive network provides a set of candidate hypothesized regions of
88 interest to account for increased pain caused by sleep deprivation. Specifically, activity
89 within regions related to the magnitude of experienced pain including somatosensory
90 cortex and thalamus should scale with the degree of enhanced pain following sleep
91 loss. In contrast, higher-order evaluative regions of the insula, cingulate, and NAcc that
92 play a role in the gating, integration, and classification of pain signals, would conversely
93 fail to appropriately modulate pain following sleep loss, evidenced by a reduction in
94 activity.

95 Beyond changes in functional brain network activity associated with acute sleep
96 deprivation, an additionally unanswered question centers on whether more subtle
97 changes in sleep from one night to the next, *within* an individual, lead to ensuing daily
98 fluctuations in pain sensitivity. This not only represents an extended test of the central
99 sleep-pain hypothesis complementing experimental acute sleep deprivation
100 manipulations, but further determines whether such findings hold public-health

101 relevance. Previous survey studies establish that sleep disruption and pain are
102 significantly linked (Afolalu et al., 2017). However, these studies typically employ cross-
103 sectional analyses at a single time point, and in clinical populations (e.g. chronic pain
104 and/or insomnia) (Tang et al., 2012). It therefore remains unknown whether micro-
105 longitudinal studies that examine changes in night-to-night sleep and day-to-day pain
106 *within* an individual are significantly and predictively inter-related. Moreover, which
107 features of sleep (e.g. duration, quality, efficiency) are the deterministic factors
108 underlying the sleep-pain relationship remains similarly unknown (Edwards et al., 2008).

109 Building on this evidence, we test a set of interrelated hypotheses: (1) acute
110 sleep deprivation increases pain-related activity within the human primary
111 somatosensory cortex and subcortical thalamus yet disrupts pain processes in regions
112 of second order limbic cortex and NAcc. Moreover, the magnitude of such changes will
113 predict the lowering in thermal pain thresholds and thus increased range of pain
114 sensation following sleep deprivation, and (2) outside of the laboratory, ecologically
115 common nightly changes of sleep within individuals in a sample of the general
116 population will result in a similar lowering of pain thresholds, expressed in daily
117 increases in experienced pain.

118

119

120 **Materials and Methods**

121 ***Experimental Design***

122 To test the experimental hypotheses, two related studies were performed: 1) an
123 in-laboratory study and, 2) an online study. The in-laboratory study involved 25 healthy
124 adult participants enrolled in a counterbalanced, repeated-measures design involving
125 two conditions: one night of sleep and one night of sleep deprivation (**Fig. 1A**). In each
126 condition, participants performed a standardized thermal pain sensitivity assessment.
127 Thermal pain thresholds were assessed outside of the fMRI scanner, with additional
128 pain-experiential questions asked during the task inside the scanner (both described
129 below).

130 In the sleep-deprived session, participants arrived at the laboratory at 10:00 PM
131 and were continuously monitored throughout the enforced waking period. Activities
132 during the sleep deprivation period were limited to use of internet, email, short walks,
133 reading, movies of low emotionality, and playing board games, thereby providing a
134 standardized regiment of waking activity without undue stress. In the sleep rested
135 session, participants came to the lab at 8:00 PM and were prepared for an
136 approximately 8 hour in-bed polysomnographic (PSG) night of sleep recording in the
137 laboratory (11:00 PM - 7:30 AM $\pm$ 30 minutes; details below). The following morning at
138 approximately 8:30 AM ($\pm$ 60 minutes), participants underwent quantitative sensory
139 testing (QST) outside of the MRI scanner to determine their pain threshold (described in
140 detail below). Thereafter, participants performed a thermal pain sensitivity task in the

141 fMRI scanner. These two sessions were separated by at least 7 days, with the order of
142 the sleep rested and deprived sessions counterbalanced across subjects.

143 In the online phase of the study, a separate cohort of 236 users of MTurk were
144 assessed across two nights and two subsequent days, sleeping as they chose (**Fig. 1B**,
145 described below). This allowed for a test of how natural night-to-night variability in sleep
146 quality, efficiency, and duration were associated with consequential day-to-day changes
147 in pain.

148 Studies were approved by the local human ethics committee, with all participants
149 providing written informed consent.

150

151 *Participants: In-laboratory study*

152 Twenty-five healthy adults, age 18 - 30 years (mean: 20.8, s.d. $\pm$ 1.95 years,
153 48% Female) completed a repeated-measures cross-over design (described above).
154 Participants abstained from caffeine and alcohol for the 72 hours before and during the
155 entire course of the study and kept a normal sleep-wake rhythm (7 - 9 hours of sleep
156 per night with sleep onset before 1:00 AM in the morning and rise time no later than
157 9:00 AM) for the 3 nights prior to the study participation, as verified by sleep logs and
158 actigraphy.

159 Exclusion criteria, assessed using a prescreening semi-structured interview
160 were: a history of sleep disorders, chronic pain or pain disorder, current pain or injuries,
161 neurologic disorders, autoimmune disorders, open and closed head injury, Axis-I
162 psychiatric disorders according to the DSM-IV-TR criteria (encompassing mental

163 disorders including depression, anxiety disorders, bipolar disorder, attention deficit
164 disorder and schizophrenia), history of drug abuse and current use of anti-depressant,
165 psycho-stimulant or hypnotic medication. Subjects who reported drinking 3 or more
166 caffeine-containing beverages a day such as caffeinated coffee, tea or soft drinks, were
167 excluded.

168 Finally, prior to their first session, participants underwent QST in order to
169 measure baseline thermal sensitivity. Two temperature levels were established for each
170 participant to be used in subsequent fMRI pain tasks in both conditions. First, a lower
171 temperature described as 'warm, but not painful', and, second, a higher temperature
172 consistently described as 'painfully hot' (corresponding to a 7/10 on a visual analogue
173 pain scale; VAS). These individualized temperatures were estimated using established
174 psychophysical methods (described below). Crucially, by matching temperatures across
175 experimental conditions, further comparisons of brain pain-reactivity controlled for
176 objective stimulus temperature. This importantly means that observed neural changes
177 caused by sleep deprivation could not be due to subjective differences in pain
178 experience elicited by objectively different stimulus temperatures.

179 To assess the degree of difference between the structured sleep schedule of the
180 experiment and each participant's unrestricted sleep schedule, participants completed
181 the Pittsburgh Sleep Quality Index (PSQI) upon study entry. This instrument contains
182 questions relating to the bed time, rise time and duration of sleep episodes across the
183 past month (Buysse et al., 1989). Additionally, in order to better characterize recent
184 sleep status, participants further completed sleep logs three days before each
185 experimental session.

186 Participants conformed to the structured sleep schedule during the month prior to
187 the experiment, including across the three days prior to the experimental session.
188 Specifically, in the month leading up to the study, participants reported average bed
189 times before 1:00 AM (mean: 12:07 AM, s.d. $\pm$ 44 minutes), rise times before 9:00 AM
190 (mean: 8:25 AM, s.d. $\pm$ 41 minutes) and sleep duration lengths between 7 and 9 hours
191 (mean: 7.63 hours, s.d. $\pm$ 0.65 hours). Similarly, for the two days prior to the start of the
192 structured sleep schedule, participants reported sleep-log mean bed times of 12:16 AM
193 (s.d. $\pm$ 40 minutes), mean rise times of 8:05 AM (s.d. $\pm$ 43 minutes) and mean sleep
194 durations of 7.71 hours (s.d. $\pm$ 0.62 hours). While it is important to note the inherent
195 limitations of self-report measures, these findings support the likelihood that participants
196 were entering the study in a rested state, and that their normative schedules were
197 congruent with the study requirements.

198 Changes in mood and anxiety state were assessed at 9 AM on the experimental
199 mornings, in both rested and deprived conditions, using the Positive and Negative
200 Affective Scale (PANAS (Watson et al., 1988) and the State-Trait Anxiety Inventory
201 (STAI (Spielberger, 1983)), respectively. As alterations in mood and anxiety commonly
202 co-occur with sleep disruption (Harvey et al., 2011), these measures provide an
203 important test of whether changes in pain experience were not due to heightened
204 anxiety and/or reduced mood.

205

206 *Thermal Sensitivity Assessments*

207 QST was performed within the scanner room, with the participant supine, by the
208 same investigator in each session. Testing was performed on the left leg, and thermal
209 stimuli were delivered using a 3 x 3 cm contact thermode consisting of Peltier elements
210 (TSA-II Neurosensory Analyzer, Medoc Ltd. Advanced Medical System, Israel). In all
211 following tests, cut-off temperature was set at 50°C.

212 Warm detection thresholds (WDT) were measured using the ascending method
213 of limits (Gracely and Naliboff, 1996). A series of 6 ascending stimuli, starting at a
214 baseline temperature of 32°C and increasing at a rate of 1°C/s, were delivered until the
215 participant terminated the stimuli with a button-press when warmth was detected, at
216 which point the temperature returned to baseline at a rate of 2°C/s. The WDT measured
217 before the first experimental session was used to determine the nonpainful warm
218 stimulus later used in the fMRI task. Specifically, 2°C were added to each participant's
219 baseline WDT and this new temperature was presented to subjects to confirm that it
220 was perceived as 'warm, but not painful'.

221 To determine the high temperature, painful stimulus for use in the fMRI task,
222 baseline suprathreshold sensitivity, a temperature corresponding to a 7/10 (with 10
223 being unbearable pain) on a VAS, was obtained for participants prior to their first
224 session. A standard staircase psychophysics method was employed in which the
225 temperature is changed on a trial by trial basis according to the participant's rating
226 (Gracely and Naliboff, 1996). In this method, a series of 8 temperatures was
227 successively applied in order to accurately estimate a consistently painful temperature.
228 Starting at the baseline temperature of 32°C, stimuli were increased at a rate of 4°C/s
229 until the target temperature was reached and held for 10 seconds. The first

230 suprathreshold test stimulus was set at 45°C for all participants, and adjusted based on
231 subsequent individual responses. Following a ramp-down at a rate of 4°C/s, participants
232 were asked to rate the intensity of the previous temperature on a VAS. Unless the
233 stimulus was rated as the target of 7 out of 10, the subsequent trial temperature was
234 adjusted by 1°C to bring it closer to the target. Once a temperature was identified as
235 having the target intensity, stimuli were held constant for subsequent trials until it was
236 rated as a 7/10 for 3 trials in a row, or the end of the stimuli set was reached, in which
237 case, the last 3 temperatures were averaged and used as the individualized painful
238 temperature in the fMRI task.

239 Thermal pain thresholds were measured in each experimental condition (sleep
240 rested, sleep deprived) using the ascending method of limits. Here, participants are
241 instructed to terminate the increasing temperature with a button press once the stimulus
242 was perceived to be unpleasant, at which point the temperature returned to baseline at
243 a rate of 4°C/s. The mean of these 6 temperatures was then used for subsequent
244 analyses. In each of the two experimental sessions, pain thresholds were assessed 30
245 minutes prior to scanning.

246

247 *fMRI Acquisition*

248 Blood oxygenation level-dependent (BOLD) contrast functional images were
249 acquired with echo-planar T2*-weighted (EPI) imaging using a Siemens 3-Tesla MRI
250 scanner with a 12-channel head coil. Each image volume consisted of 37 descending
251 3.5mm slices (96 x 96 matrix; TR = 2000ms; TE = 22ms; size 3.5 x 3.5 x 3.0mm, flip

angle = 50°, 0.5mm slice spacing). A high-resolution T1 weighted structural scan was acquired at the end of the sleep rested session (256 x 256 matrix; TR=1900; TE = 2.52; flip angle = 9°; FOV 256mm; 1 x 1 x 1mm voxels).

fMRI Thermal Pain Sensitivity Task

Each scanning session for the thermal pain sensitivity task consisted of a pseudo-randomly ordered sequence of two stimulus intensity phases: painful hot, and nonpainful warm. Twelve painful and twelve nonpainful warm stimulus 'trials' (twenty-four, total) were presented across three scanning blocks in each of the sleep-rested and sleep-deprived conditions. Each block was separated by at least 60 seconds, with each of the three blocks lasting approximately 5 minutes.

For each of the 24 experimental trials per session, participants were presented with a visual cue signaling impending thermal stimulation lasting 6 seconds, after which the cue was removed from the screen and replaced with a fixation cross, followed by painful/nonpainful stimulation. The stimulation period consisted of a ramp-up period in which heat increased at a rate of 4°C/s and was held at the target temperature for 12 seconds. Consistent with prior fMRI studies of neural pain assessments (Becerra et al., 1999; Brown et al., 2011; Zunhammer et al., 2015), only the 12-second period of peak temperature stimulation was used in modeling of brain activity during painful/nonpainful trials. The heat returned to baseline (32°C) at a rate of 8°C/s, after which an 8 second period of relief from stimulation occurred. Finally, after each stimulation trial, subjects rated both the intensity and unpleasantness of the previous temperature on an 11-point

274 scale. All trial durations were jittered according to a uniform distribution, producing inter-
275 trial interval variability that is optimal for mixed block/event-related designs (Petersen
276 and Dubis, 2012).

277

278 *Online Mechanical Turk Study*

279 The online phase of the study was conducted through Amazon Mechanical Turk.
280 We sought to assess a cohort of at least 50 participants who reported experiencing
281 some current pain, thereby allowing a determination of whether night-to-night changes
282 in sleep predicted subsequent day-to-day changes in their pain experience. All
283 participants were queried at 4 time points across 2 nights and days. In the mornings,
284 participants reported on the prior nights' sleep, and in the evenings, reported on the
285 days' pain.

286 A total of two-hundred and thirty-six (mean age 36.61, s.d. 10.8 years; 50%
287 Female) participants were surveyed. Only participants who reported: pain on one or
288 both assessment days, an IP address in the United States, and a prior online MTurk
289 approval rating of 95% or higher, were included in subsequent analyses. Based on
290 these factors, a subset of 60 participants (mean age 38.4, s.d. 10.4 years, 42% female)
291 of candidates surveyed were eligible for participation (**Fig. 1B**).

292 Across the experimental phase, participants quantified a number of factors
293 related to sleep and pain within specified time windows of the mornings and evenings.
294 In morning assessments, participants filled out the Karolinska sleep diary, which
295 included bed time, sleep latency, wake-up time, sleep duration, and subjective sleep

296 quality on a 5-point scale (Åkerstedt et al., 1994). Sleep efficiency was calculated from
297 these daily diaries, based on percent of time asleep out of total sleep duration (i.e. total
298 sleep time minus sleep latency and total wake time after sleep onset). In order to
299 characterize changes in affect, participants reported current mood and anxiety, using
300 the PANAS and STAI questionnaires, respectively.

301 To assess the influence of nightly sleep on subsequent next-day pain,
302 participants quantified their physical pain experience that day (inputted between 6PM -
303 day's end) and rated the average magnitude of that pain on a 100-point scale (0
304 representing 'No Pain' and 100 representing 'Worst Pain Imaginable').

305 Similar to the in-lab experiment, the online study employed a within-subject,
306 repeated measures design. This minimized differences beyond the variable of sleep that
307 could otherwise influence the outcome measure of pain (e.g. sex, age, health status).
308 We deliberately did not provide instructions to subjects to curtail or elongate their sleep
309 in order to maintain ecologically naturalistic sleep changes. To maximize the
310 representativeness of the online sample, participants were deliberately not excluded
311 based on age or health status, including the presence of chronic pain or pain disorders.
312 Data points for either sleep or pain metrics that were ± 2 s.d. of the group mean were
313 considered outliers and excluded from analyses.

314

315

316 ***Statistical Analyses***317 *fMRI Preprocessing*

318 Preprocessing and data analysis were performed using the Statistical Parametric
 319 Mapping ([SPM12](#), RRID:SCR_007037) software implemented in Matlab R2015b
 320 ([MATLAB](#), RRID:SCR_001622). Functional images were motion corrected and slice
 321 time corrected. Structural images were then coregistered to the mean functional image
 322 and normalized to a standard template. Functional images were spatially normalized to
 323 the Montreal Neurological Institute template using those parameters and smoothed
 324 using an 6mm full-width-at-half-maximum (FWHM) Gaussian kernel. For all pre-
 325 processing steps, we used the default parameters in SPM12. For each subject, trial-
 326 related activity was assessed by convolving a vector of trial onsets with a canonical
 327 hemodynamic response function. The six movement-related covariates (three rigid-body
 328 translations and three rotations determined from the realignment preprocessing step)
 329 were used as regressors in the design matrix for modeling movement related artifact in
 330 the time series. To further address the influence of motion on BOLD data, frame-wise
 331 displacement (FD) of head motion was calculated based on the motion parameters
 332 estimated during preprocessing using the ArtRepair toolbox ([ArtRepair for robust fMRI](#),
 333 RRID:SCR_005990). TRs with FD values larger than 0.8 were interpolated with the
 334 nearest artifact-free TRs surrounding the time-point effected by motion. Autocorrelation
 335 in the data was accommodated for using a first order autoregressive (AR1) model,
 336 estimated by pooling over suprathresholds (Friston et al., 2002).

337 To control for physiological noise, 5 principal components of cerebrospinal fluid
 338 (CSF) and white matter (WM) signal were added as regressors to the design matrix,

339 implemented through the CompCor pipeline (Behzadi et al., 2007). Extraction of
340 CSF/WM signal was derived from probabilistic maps segmented from T1-weighted
341 structural images of each participant, using the segment function within SPM12.
342 CSF/WM masks were then threshold at a probability of 0.99 for WM and 0.95 for CSF,
343 converted to functional-scan resolution and eroded to eliminate isolated voxels and
344 minimize partial volume effects.

345

346 *fMRI analyses*

347 A general linear model (GLM) was specified for each participant, using the task-
348 related and nuisance regressors outlined above, to investigate the neural effects of
349 interest (Friston et al., 1994). Contrasts were created at the first (individual) level with
350 two events: painful and nonpainful heat. Specifically, the 12 second period during which
351 the stimulus was at its peak temperature, excluding temperature ramp-up and ramp-
352 down periods, was used to construct these contrasts. The experimental hypotheses
353 were tested at the second (group) level using a priori regions of interest (ROI) centered
354 on previously reported coordinates. Averaged activity from the current study was
355 extracted from these

356 ROI's and further analyses with correction for multiple statistical tests were
357 performed on these values. The approach prevents analysis circularity arising from non-
358 independence of the selected regions and the results, and is in accord with
359 standardized fMRI methods (Poldrack, 2007; Poldrack and Mumford, 2009).

360 A set of cortical and sub-cortical regions of interest (ROI's) were defined on the
361 basis of three convergent criteria. First, regions were selected from previously reported
362 fMRI patterns associated with objective pain intensity. These regions were derived from
363 machine-learning analysis of fMRI brain activity during thermal pain stimulation, and
364 thus, represent a set of candidate neural regions whose activity reliably relates to
365 objective noxious stimulus intensity (Wager et al., 2013). Within this network, only those
366 regions in which increased activity was independently and significantly ($q < 0.05$ FDR)
367 related to the painful stimuli were considered in the current ROI analysis.

368 Second, regions contributing to subjective pain experience, but not directly
369 related to objective stimulus intensity, were considered. These regions were derived
370 from prior findings of fMRI brain activity associated with subjective thermal pain
371 experience (Woo et al., 2017). This was important in the context of the current
372 experiment since prior studies of sleep loss have shown changes in subjective
373 estimates of pain in response to the same objective noxious stimulus intensity (Tiede et
374 al., 2010; Campbell et al., 2011). Here again, only those regions in which activity
375 independently and significantly ($q < 0.05$) contributed to subjective pain were included
376 as candidate ROI's.

377 Third, affective processes participate in painful experiences (Price, 2000; Wiech
378 and Tracey, 2009) and this may contribute to the hyperalgesic effect of sleep
379 deprivation. We therefore additionally required that a priori pain-related regions of
380 interest were known to be sleep-loss sensitive in prior studies of basic affective and
381 emotional neural processing (Venkatraman et al., 2007; Yoo et al., 2007; Gujar et al.,
382 2011; Goldstein et al., 2013; Greer et al., 2013; Goldstein-Piekarski et al., 2015).

383 From these three criteria, a set of 15 candidate regions of interest emerge,
384 reflecting the Venn diagram overlap of these previously published empirical studies:
385 areas that are conjointly involved in objective noxious stimulus processing, subjective
386 pain sensation, and sensitive to the effects of sleep deprivation in affective tasks.
387 Sphere sizes of the ROI's used in the current study were selected to approximate the
388 cluster sizes (mm^3) reported in the original studies from which coordinates were taken
389 (**Table 1**).

390

391 *Pain and Brain Analysis*

392 To characterize the impact of sleep deprivation on the pain-related ROIs,
393 average contrast estimates (Pain Vs. No Pain) were extracted from each ROI volume
394 independently and condition differences were calculated using two-tailed paired t-tests.
395 All statistical tests were performed within MATLAB. Consistent with imaging
396 recommendations to correct for multiple statistical comparisons across the 15 ROIs, the
397 Benjamini-Hochberg procedure for controlling false discovery rate (FDR) was applied,
398 with $q = 0.05$ (Benjamini and Hochberg, 1995). This procedure addresses the problem
399 of simultaneously performing multiple significance tests by using an adaptive stepwise
400 procedure to control the expected ratio of erroneous rejections to the number of total
401 rejections. P-values which survived the FDR criterion after correction were considered
402 statistically significant.

403 In order to test the relationships between sleep-loss-related changes in ROI brain
404 activity and subjective pain reports, percent-change in pain thresholds, and

405 suprathreshold pain (intensity and unpleasantness) were calculated for each subject,
406 and Pearson correlated to changes in pain ROIs. The same FDR approach described
407 above was used to correct for multiple tests.

408

409 *Mechanical Turk Analysis*

410 Day-to-day (day2 – day1) changes in reported pain, and night-to-night (night2 –
411 night1) changes in sleep were calculated for each participant. Testing the hypothesis
412 that natural fluctuations in three measures of sleep predicted subsequent next-day pain,
413 participants were categorically separated on the basis of whether they experienced an
414 increase or decrease from one night to the next in each of the three sleep measures
415 (above/below 0% change). Of the 60 participants, 55% improved their sleep quality from
416 one night to the next. Forty-percent of subjects improved their sleep quantity/duration,
417 and 52% improved their sleep efficiency relative to the first assessed night. Daily
418 changes in reported pain were compared between categorical groups, defined on the
419 basis of each of the three measured sleep factors, using two-tailed paired t-tests (*alpha*
420 = 0.05) (**Fig. 5**).

421

422

423 **Results**

424 *Sleep deprivation and pain sensitivity*

425 In-laboratory study: Consistent with the hypothesis, the standardized thermal
426 pain threshold assessment (performed outside of the MRI scanner) demonstrated a
427 significant impact of sleep deprivation on pain processing. Specifically, sleep loss
428 expanded the temperature range for classifying a stimulus as painful through a lowering
429 of pain threshold, such that sleep deprived participants registered pain at a significantly
430 lower temperature than when sleep rested ($t[24] = 4.36$, $p = 0.0002$; **Fig. 2**).

431 Inside of the scanner, there were no significant main effects of sleep deprivation
432 on ratings of intensity and unpleasantness of supra-threshold noxious blocks (Intensity:
433 $t[24] = 0.67$, $p = 0.51$; Unpleasantness: $t[24] = 0.94$, $p = 0.36$). This was similarly true for
434 ratings of nonpainful stimulus blocks (warmth) (Both: $t[24] < 0.98$, $p > 0.33$). This
435 importantly demonstrates that sleep loss did not significantly alter sensation of
436 unambiguous non-noxious thermal stimuli.

437 Therefore, sleep deprivation expressly resulted in a shift—specifically a
438 lowering—in the decision threshold to classify a stimulus as painful (assessed outside of
439 the scanner), as shown by the significant decrease in pain thresholds, while the
440 extremes of pain sensitivity used in the canonical scanning paradigm remained similar
441 in each condition. The latter importantly meant that any differences in brain activity
442 between the sleep rested and sleep deprivation conditions assessed during fMRI
443 scanning could not simply be accounted for by one condition having an imbalance in
444 subjective experience ratings.

445

446 *Sleep deprivation and neural pain processing*

447 Analyses of fMRI data revealed bidirectional changes in brain activity in
448 participants under conditions of sleep loss within recognized *a priori* pain regions,
449 relative to when sleep rested. Specifically, sleep deprivation significantly *increased* pain
450 reactivity within right primary somatosensory cortex ($t[24] = 3.20$, $p = 0.004$; false
451 discovery rate (FDR) corrected for multiple ROI tests at $q = 0.05$ (Benjamini and
452 Hochberg, 1995); **Fig. 3**). This finding is consistent both with the role of somatosensory
453 cortex in the registration and representation of noxious painful stimuli (including heat),
454 and the contralateral crossing of sensory signals, since the stimulation was to the left
455 side of the body (Coghill et al., 1999; Craig, 2002; Apkarian et al., 2005; Atlas et al.,
456 2014).

457 In contrast to the somatosensory cortex, significant *decreases* in activity were
458 observed within subcortical *a priori* ROIs of the thalamus as well as the NAcc following
459 sleep deprivation, relative to the rested condition (Thalamus: $t[24] = -3.13$, $p = 0.005$;
460 NAcc: $t[24] = -3.11$, $p = 0.005$; FDR corrected). Such NAcc disengagement is congruent
461 with the prediction of a shift in decision threshold regarding what constitutes a painful
462 stimuli, since this striatal region is associated with pain valuation (Altier and Stewart,
463 1999; Baliki et al., 2010; Navratilova and Porreca, 2014; Lee et al., 2015; Woo et al.,
464 2015; Woo et al., 2017; Sardi et al., 2018). Appropriate cortical estimation of pain within
465 the human brain is not limited to representations within primary somatosensory strip, but
466 involves further higher-order cortical evaluation within interoceptive integration regions
467 of the insula and cingulate (Craig, 2003; Critchley et al., 2004). *A priori* ROI regions of

the right middle insula and left anterior insula demonstrated significant decreases in activity during pain under conditions of sleep deprivation (Right Middle Insula: $t[24] = -4.06$, $p = 0.0005$; Left Anterior Insula: $t[24] = -3.11$, $p = 0.005$; FDR corrected). The latter, left hemisphere association is relevant considering that left anterior insula receives parasympathetic afferent information, and lesions of this insula region increase pain sensitivity--indicative of a loss of anodyne homeostatic signaling under sleep-deprivation conditions (Craig, 2005; Starr et al., 2009).

No other regions within the *a priori* cortical pain network demonstrated significance that exceeded FDR correction for multiple ROIs, including the cingulate cortex. Several were significant below this threshold, and reported for completeness in **Table 1**.

Having identified condition differences (sleep rested < > sleep deprived) in pain-related brain activity, we next sought to determine whether the *extent* of change in activity within these pain-related regions predicted inter-individual differences in the change in pain threshold caused by sleep deprivation.

Across participants, the extent of amplified somatosensory pain reactivity positively and significantly predicted the lowering of pain thresholds (i.e. the expansion of pain sensation) caused by sleep deprivation ($r(23) = 0.55$, $p = 0.004$; FDR corrected; **Fig. 4A**). In addition, the degree of decrease in thalamic activity associated with sleep deprivation significantly and negatively predicted the lowered of pain thresholds across individuals ($r(23) = -0.54$, $p = 0.005$; FDR corrected; **Fig. 4B**). That is, the extent of increased somatosensory pain reactivity, and the associated decrease in thalamic reactivity, following sleep deprivation, both predicted increasingly larger sleep-loss

491 related changes in pain threshold across individuals. No other such associations
492 (positive or negative) were identified within other *a priori* ROIs (all $p > 0.05$; **Table 1**).

493 Therefore, the somatosensory cortex and left thalamus both showed a main
494 effect of sleep deprivation, and furthermore, their respective extent of change in pain-
495 reactivity predicted the magnitude of expanded sensitivity to pain caused by sleep
496 deprivation, measured using the threshold assessment.

497 Mood and anxiety are known to be impacted by sleep deprivation (Harvey et al.,
498 2011), and independent of sleep, are recognized to influence pain (Wiech and Tracey,
499 2009). In order to test the specificity of the pain-amplifying effect of sleep deprivation,
500 relationships between self-reported changes in mood and anxiety following sleep
501 deprivation were examined in relation to the change in pain.

502 Levels of both positive and negative mood, reported immediately before
503 scanning, were numerically but not significantly different between sleep deprived and
504 rested conditions (Positive Mood: Sleep deprived mean = 17.4 [± 4.8 s.d.]; Sleep rested
505 mean = 20.12 [± 7.4]; $p = 0.14$; Negative Mood: Sleep deprived mean = 15.2 [± 1.9 s.d.];
506 Sleep rested mean = 16.64 [± 4.8]; $p = 0.19$). Importantly, however, these sleep-loss
507 related differences in both positive and negative mood did not significantly correlate with
508 the change in pain thresholds (Positive Mood: $r(23) = -0.04$, $p = 0.85$; Negative Mood:
509 $r(23) = -0.18$, $p = 0.39$). There were similar numeric but non-significant differences in
510 state anxiety following sleep deprivation. However, like mood, these sleep-loss
511 associated changes in self-reported anxiety were also not correlated with changes in
512 pain thresholds measured at the same time (Anxiety: Sleep deprived mean = 35.5
513 [± 10.2 s.d.]; Sleep rested mean = 32 [± 9.3 s.d.]; $p = .18$; $r(23) = -0.27$, $p = 0.19$).

514 Taken together, this suggests that the observed associations between sleep and
515 pain were not parsimoniously accounted for by co-occurring changes in affect in the
516 current study.

517

518 *Ecological night-to-night changes in sleep and subsequent day-to-day changes in pain*

519 While the in-laboratory experiment establishes that acute total sleep deprivation
520 leads to altered neural pain processing and an associated expansion of sensitivity to
521 pain, it predicts, but does not itself address, whether ecologically relevant night-to-night
522 variation in sleep, within an individual, similarly results in consequential day-to-day
523 changes in experienced pain during the subsequent day.

524 Supporting the experimental hypothesis, and the directional predictions from the
525 in-lab acute sleep deprivation experiment, participants who had a decrease in sleep
526 efficiency from one night to the next experienced a corresponding increase in pain from
527 one day to the next ($t[58] = 2.4$, $p = 0.018$; **Fig. 5**; each bar representing a different
528 group of subjects).

529 A similar relationship was observed for night-to-night changes in subjective sleep
530 quality, such that reduction in the quality of sleep was associated with increased
531 consequential daily pain ($t[58] = 2.1$, $p = 0.04$; **Fig. 5**). Interestingly, however, nightly
532 changes in total amount of sleep did not similarly predict daily changes in pain ($t[58] =$
533 0.39 , $p = 0.7$).

534 Though the current study was not designed to prove a causal order of changes in
535 sleep and changes in pain, the data suggest that altered sleep statistically forecasts

536 next-day pain. To further examine if this were the case, we used partial correlation
537 analysis to examine the relative strength of the association between nightly sleep quality
538 and next-day pain, while controlling for effects of pain on the previous day. If sleep
539 quality still predicted next-day pain, even when controlling for prior pain status, it would
540 lend support to the idea that it is the change in sleep that is robustly predicting
541 subsequent changes in pain. Additionally, we extend this test and performed a second
542 partial correlation analysis of pain and next-day sleep quality while controlling for prior
543 sleep status. This affords an additional exploration that speaks to causal precedence;
544 were pain to be associated with subsequent sleep quality independent of the effects of
545 prior sleep, it would suggest that pain changes *precede* sleep changes, not vice versa.

546 When controlling for the effects of pain intensity on day 1, a significant
547 relationship between sleep quality and pain on the following day (day 2) still remained (r
548 = -0.26, $p = 0.04$). This suggests that the evening sleep quality is a stronger predictor of
549 subsequent next-day pain than prior pain predicting subsequent sleep. In addition, when
550 controlling for prior sleep quality (night 1), the partial correlation between pain and sleep
551 quality on the following night (night 2) was non-significant ($r = -0.04$, $p = 0.72$). These
552 results therefore affirm that it is sleep quality that most robustly predicts next-day pain,
553 even when factoring in the influence of prior pain status.

554 Importantly, self-reported changes in mood and anxiety, were not associated with
555 day-to-day fluctuations in experienced pain (Positive Mood: $r(58) = -0.14$, $p = 0.29$;
556 Negative Mood: $r(58) = -0.21$, $p = 0.11$; Anxiety: $r(58) = 0.07$, $p = 0.59$). As with the in-
557 laboratory study, this would indicate that the observed associations between sleep and

558 pain do not appear to be robustly explained by parallel changes in these emotional
559 constructs.

560 Thus, ecological night-to-night variability in sleep efficiency and subjective quality
561 within an individual, but not quantity, predicted consequential day-to-day reductions in
562 corresponding pain. This online counterpart to the in-lab experiment corroborates the
563 pain enhancing effect of inadequate sleep, but further establishes that even modest
564 changes in sleep quality are sufficient to affect pain, and can be observed within
565 individuals across nights, beyond cross-sectional examinations.

566

567 **Discussion**

568 Taken together, these findings help establish 1) that acute sleep deprivation
569 expands the temperature range for classifying a stimulus as painful, specifically through
570 a lowering of pain thresholds, 2) that the underlying neural correlates of such sleep-loss
571 induced hyperalgesia involved increase pain reactivity with primary somatosensory
572 cortex, yet a blunting of pain-reactivity in higher-order valuation and decision-making
573 regions of the striatum and insula cortex, and 3) the same effect can be observed
574 following more ecologically relevant changes in sleep, such that modest night-to-night
575 changes in sleep quality within an individual (increases and decreases) determine
576 consequential day-to-day changes in experienced pain (decreases and increases,
577 respectively).

578

579 *Sleep and Neural Pain Processing*

580 Current models view pain as a homeostatic emotion. Pain is constructed from
581 ascending interoceptive afferents conveying viscerosensory and viscerosensory
582 changes in body state. These signals project to registration sites within brainstem, and
583 on to subcortical and cortical sensory, homeostatic regulatory, and valuation centers.
584 Pain is therefore a specific sensation and a felt emotion, one that drives homeostatic
585 behavior (Craig, 2002, 2003; Critchley and Harrison, 2013).

586 Fitting this distributed model of pain processing, our findings describe a
587 bidirectional set of neural changes within the recognized network of the brain that
588 processes pain following sleep deprivation. Specifically, sleep loss triggered an increase
589 in pain reactivity within sensory-discriminative regions of somatosensory cortex.
590 Notably, heightened somatosensory cortex reactivity was lateralized to the opposite
591 side of the cortex, relative to the application of the noxious stimulus on the body. This is
592 fitting with the contralateral organization of primary somatosensation, and indicates an
593 intensification of first-order cortical pain registration resulting from sleep deprivation
594 (Craig, 2002).

595 Beyond this main condition effect, the magnitude of increased activity in
596 somatosensory cortex following sleep deprivation predicted the relative decrease in
597 thermal pain threshold in the sleep-deprived state—the greater the increase in
598 somatosensory cortex noxious reactivity, the greater the shift (lowering) of pain
599 sensitivity across individuals. That the somatosensory cortex explained individual
600 differences in pain thresholds fits with the role of somatosensory cortex in the initial
601 magnitude registration of pain (Coghill et al., 1999; Apkarian et al., 2005; Atlas et al.,
602 2014). Therefore, one component of the exacerbation of pain caused by a lack of sleep

603 involves amplified primary pain processing in rudimentary areas of human sensory
604 cortex, the activity of which scales with sleep-loss induced expansion of pain sensation.

605 A converse decrease in pain reactivity was observed in subcortical NAcc,
606 thalamus, and insula cortex. A recognized function of the insula involves the integration
607 of afferent signals from somatosensory cortex into second-order representation of
608 affective body state, available for conscious perception (Craig, 2002, 2003; Critchley
609 and Harrison, 2013). Indeed, the broad connectivity of the insula with regions including
610 the prefrontal and orbitofrontal cortices, thalamus, ACC, and NAcc, suggests that the
611 insula is well-situated to be a key regulator of sensed pain (Craig, 2002, 2003).

612 Related, the NAcc is linked with pain valuation, decision-making, and self-
613 regulation in a number of studies (Navratilova and Porreca, 2014; Woo et al., 2015;
614 Woo et al., 2017). This includes NAcc-associated relief from ongoing pain, the analgesic
615 influence of which is transacted through descending pain efferent pathways (Altier and
616 Stewart, 1999; Baliki et al., 2010; Lee et al., 2015; Woo et al., 2015; Woo et al., 2017;
617 Sardi et al., 2018). In addition, the NAcc has reciprocal connections to pain-integrating
618 and pain modulating cortical regions, including ventromedial prefrontal cortex, ACC,
619 insula, and somatosensory cortex (Borsook et al., 2010).

620 The thalamus is consistently activated in acute pain experiments, and plays a
621 role in the gating of contralateral ascending sensory information as it reaches the cortex
622 (Craig, 2003). However, the function of the thalamus in pain processing is not limited to
623 classical sensory gating. First, decreases in thalamic grey matter structure and
624 functional activity are a characteristic of multiple chronic pain conditions with different
625 etiologies (Pridmore et al., 2003; Apkarian et al., 2004; Gustin et al., 2011). Indeed, the

626 reduction of thalamic function due to lesions predicts the severity of central pain
627 conditions, including reduced pain thresholds (Vartiainen et al., 2016), the proposed
628 mechanism of which is the loss, or disinhibition, of normal thalamocortical pain signaling
629 (Craig et al., 1996; Craig, 2000; Vartiainen et al., 2016).

630 Building on this functional circuitry, increased somatosensory, yet reduced insula,
631 thalamus, and NAcc, pain reactivity challenges a rudimentary model in which sleep-loss
632 triggers hyperalgesia through unidirectional increases in neural pain representations.
633 Instead, the impact of insufficient sleep on pain involves both an amplification of primary
634 cortical pain registration, potentially due to thalamic disinhibition, and a shift in affective
635 valuation and decision-making involving insula and NAcc. These changes can
636 parsimoniously account for the observed shift in the decision threshold for categorizing
637 a potentially noxious stimulus as painful through misrepresentation of pain signals within
638 insula and NAcc. This misrepresentation of the salience of thermal stimuli may
639 consequentially lead to an inaccurate amplification (gain increase) of pain registration
640 within somatosensory cortex. In fact, this exact pattern has been reported in patients
641 with lesions to the insula cortex, who demonstrate simultaneously higher pain
642 sensitivity, elevated somatosensory cortex activity, and an absence of pain-modulatory
643 activity in the insula (Starr et al., 2009).

644

645 *Ecological Variations in Sleep and Pain*

646 Further testing our hypothesis, the second study demonstrated that subtle,
647 ecological changes in sleep in the general population predict consequential day-to-day

648 changes in self-reported pain intensity. Interestingly, it was night-to-night changes in
649 sleep quality, rather than sleep quantity, that were deterministic of changes in pain
650 sensitivity.

651 Insufficient sleep, self-reported, has been shown to correlate, cross-sectionally,
652 with self-reported pain in a sample of the general population (Smith and
653 Haythornthwaite, 2004; Afolalu et al., 2017) and clinical cohorts (Tang et al., 2012). Our
654 findings add to this link by establishing that, beyond cross-sectional associations,
655 changes in the quality of night-to-night sleep *within* an individual in a micro-longitudinal
656 assay, more so than the quantity, predict consequential day-to-day changes in pain
657 intensity (Edwards et al., 2008). That even modest reductions in sleep quantity/quality
658 impact next-day pain is increasingly relevant given the continued erosion of sleep time
659 in developed nations (National Sleep Foundation Sleep in America Poll, 2013,
660 www.sleepfoundation.org/2013poll). This is additionally pertinent when considered
661 alongside the escalating prevalence and economic health burden of pain in these same
662 countries (Goldberg and McGee, 2011). Indeed, findings from a recent national poll
663 (National Sleep Foundation Sleep in America Poll, 2015,
664 www.sleepfoundation.org/sleep-polls-data/2015-sleep-and-pain) demonstrate that 64%
665 of those suffering chronic pain, and 54% of those with acute pain, report co-occurring
666 poor sleep quality.

667 Our findings perhaps offer encouragement, in that they suggest that even modest
668 *improvements* in sleep quality have the potential to reduce subjectively significant pain.
669 That is, sleep can be seen as a modifiable risk factor and intervention target for pain,

670 and the magnitude of sleep improvement necessary to result in measurable reductions
671 in pain is modest, and thus pragmatically realistic.

672 Given that pain is among the costliest health burdens in both direct healthcare
673 spending and indirect lost productivity (Phillips, 2009), improving sleep may additionally
674 confer broader economic savings. This concern (and therapeutic opportunity) is
675 exemplified by the in-patient hospital setting, where sleep quality is consistently poor
676 and pain is frequently co-occurring. Placing sleep quality closer to the center of in-
677 patient treatment approaches may reduce patient suffering, and potentially, the dosage
678 of narcotic and non-narcotic analgesics (Walker, 2018).

679

680 *Study Considerations*

681 Potential limitations of the current studies should be noted. First, the in-laboratory
682 experiment was performed with healthy young adults. Since pain becomes more
683 frequent in middle and late-life, it remains to be determined whether these findings
684 translate across the lifespan. Second, only noxious heat was examined in the laboratory
685 study. It therefore remains unclear whether different brain changes would be observed
686 using different stimulus modalities, such as noxious mechanical, electrical, or chemical
687 stimuli. This seems less likely, however, considering that previous studies show a
688 commonality of pain-enhancing effects of sleep loss across such modalities
689 (Lautenbacher et al., 2006). Third, as brain activity was only collected during the supra-
690 threshold pain sensitivity task, inferences regarding the mechanisms of reduced
691 thresholds rely on significant correlations with subsequent supra-threshold neural pain

692 reactivity. Finally, while post-hoc analyses established that changes in nightly sleep
693 statistically forecast next-day pain changes, even when accounting for prior pain status,
694 they do not prove this fact causally (as this experiment was not designed to directly
695 address causal ordering). Nevertheless, a meta-analysis of longitudinal and micro-
696 longitudinal studies in both clinical and non-clinical populations has established that
697 two-thirds of relevant studies demonstrate stronger evidence supporting a causal
698 influence of prior sleep on subsequent experienced pain, rather than the opposite
699 direction (Finan et al., 2013).

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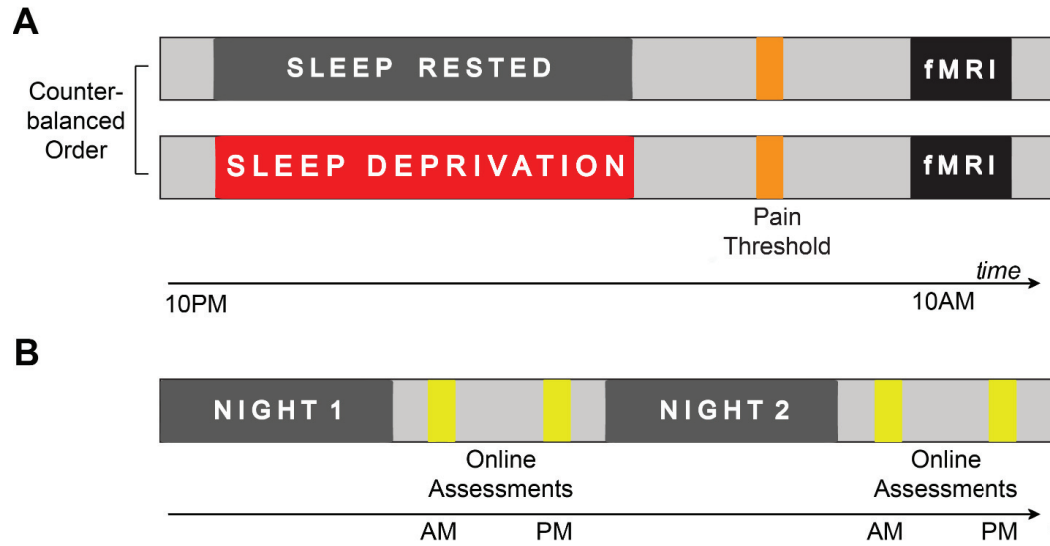
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Fig. 1. Experimental design. (A) Design of the in-laboratory repeated measures counterbalanced study involving one night of sleep deprivation and one night of sleep, followed by an fMRI scanning session involving a pain-evoking task. (B) Design of the online study involving daily sleep diaries for two consecutive nights tracking habitual variations in sleep time, followed later by daily pain assessment.

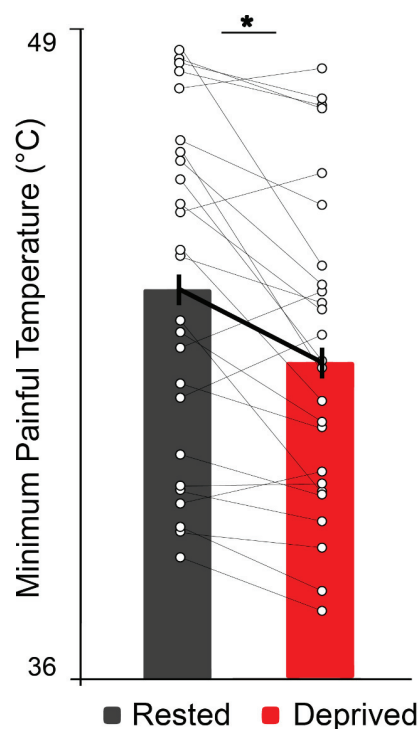


Fig. 2. Thermal pain threshold. Following sleep deprivation, there was an increase in pain sensitivity demonstrated by a significant lowering of pain thresholds, relative to sleep rested. (Sleep deprived mean = 42.47°C [$\pm 3.22^\circ\text{C}$ s.d.]; Sleep rested mean = 43.89°C [$\pm 3.49^\circ\text{C}$ s.d.]. $*P < 0.0002$).

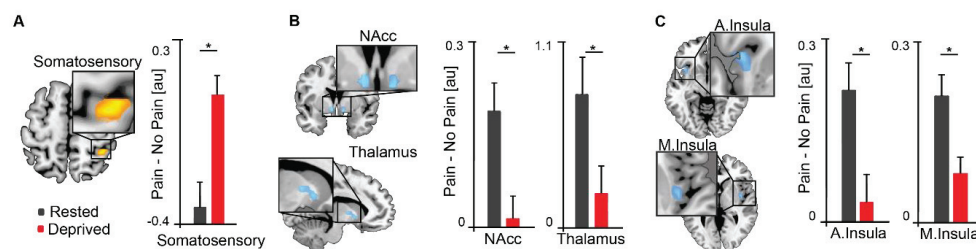


Fig. 3. Changes in pain-related brain activity. (A-C) Brain maps (left)

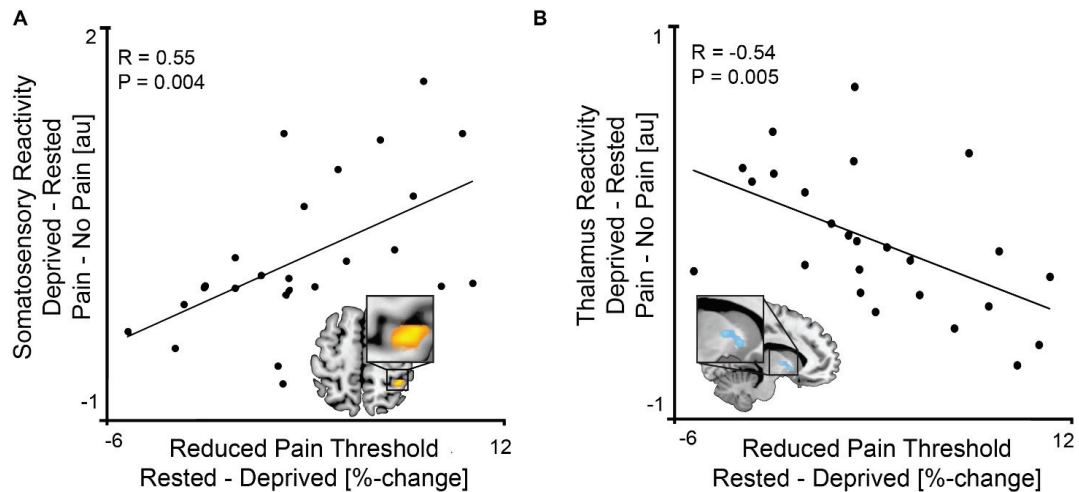
displaying pain-related regions of interest (ROIs) that were significantly altered by sleep deprivation, relative to sleep rested (all FDR corrected).

Bar plots display brain activity (Pain > No Pain) contrast estimates.

Somatosensory: Sleep deprived mean = $0.09 [\pm 0.34 \text{ s.d.}]$; Sleep rested mean = $-0.34 [\pm 0.46 \text{ s.d.}]$. Nucleus Accumbens: Sleep deprived mean = $0.02 [\pm 0.18 \text{ s.d.}]$; Sleep rested mean = $0.19 [\pm 0.21 \text{ s.d.}]$. Thalamus: Sleep deprived mean = $-0.21 [\pm 0.81 \text{ s.d.}]$; Sleep rested mean = $0.79 [\pm 1.06 \text{ s.d.}]$. Anterior Insula: Sleep deprived mean = $0.12 [\pm 0.82 \text{ s.d.}]$; Sleep rested mean = $0.79 [\pm 0.8 \text{ s.d.}]$. Middle Insula: Sleep deprived mean = $0.08 [\pm 0.12 \text{ s.d.}]$; Sleep rested mean = $0.21 [\pm 0.17 \text{ s.d.}]$. Error bars represent SEM.

* $P < 0.005$. Nucleus Accumbens (NAcc); Anterior and Middle Insula (A. Insula and M. Insula, respectively). Threshold for display set to $P < 0.005$, whole-brain, uncorrected.

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Fig. 4. Pain reactivity and subjective pain sensitivity. Scatterplot displaying significant positive correlation between sleep-loss related increases in **(A)** somatosensory and thalamic **(B)** pain reactivity (Pain > No Pain), and the sleep-loss related increases in subjective pain sensitivity (change in pain threshold). Threshold for display set to $P < 0.005$, whole-brain, uncorrected.

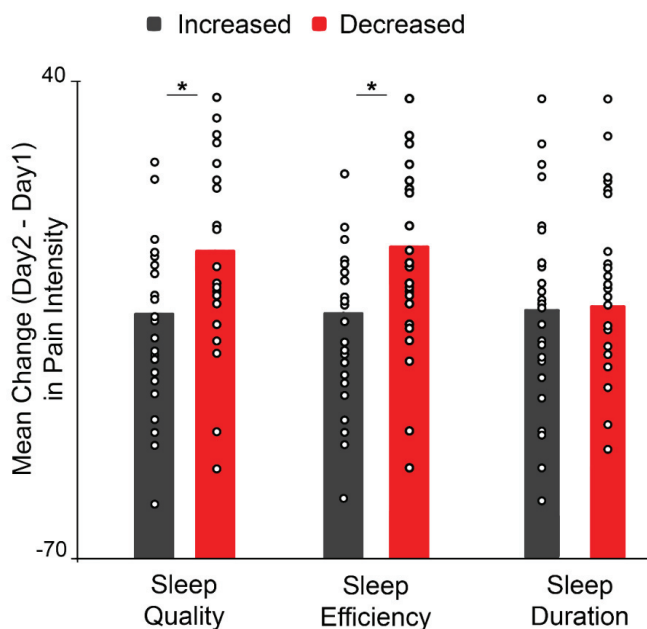
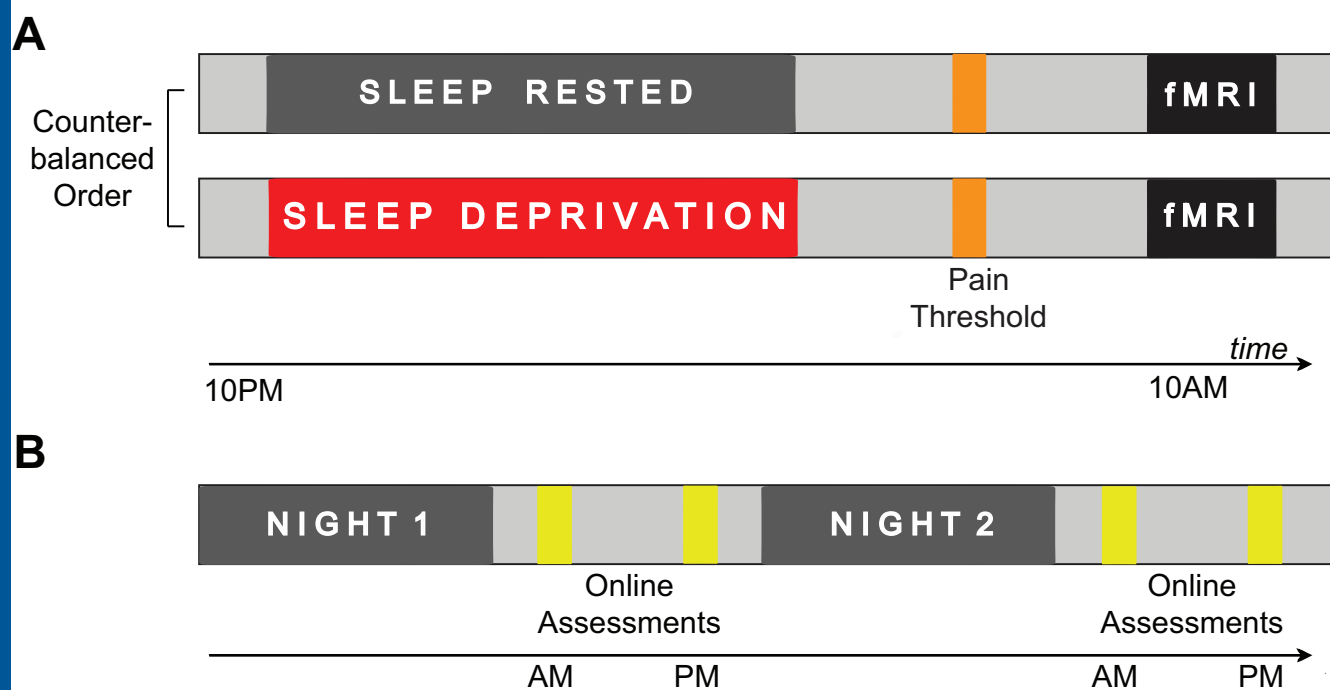
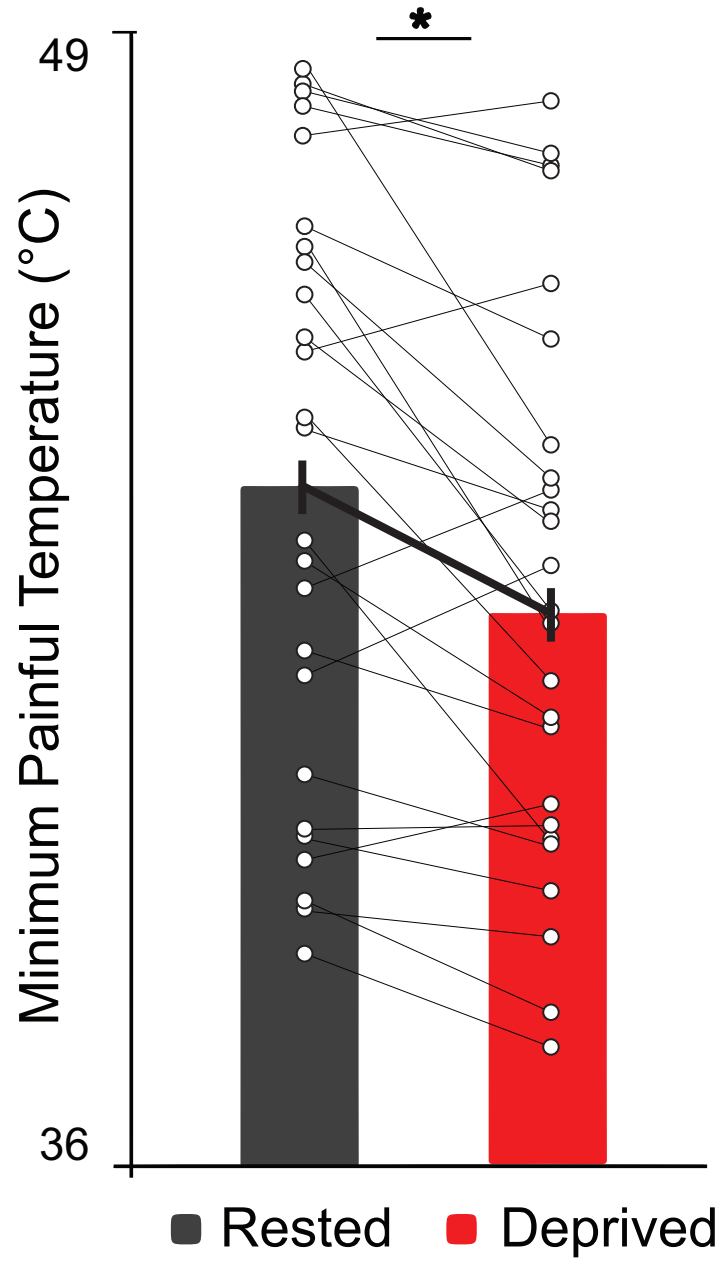
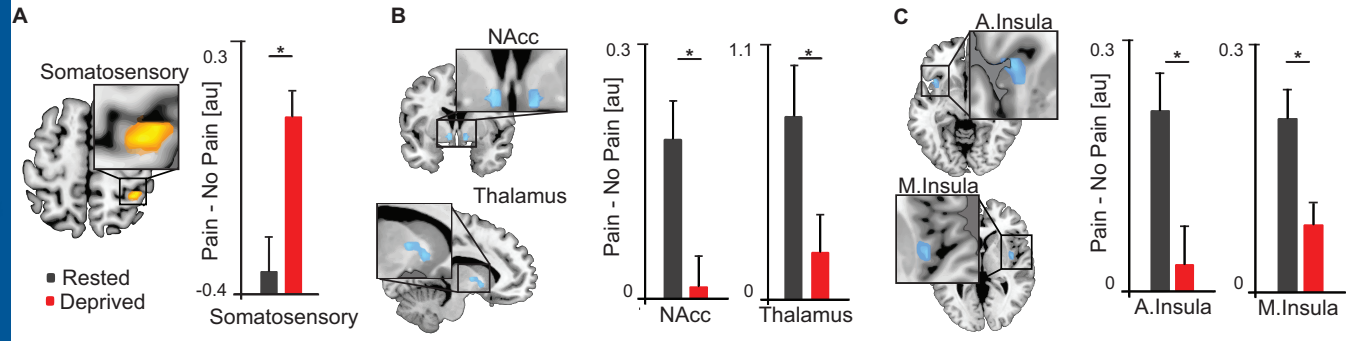
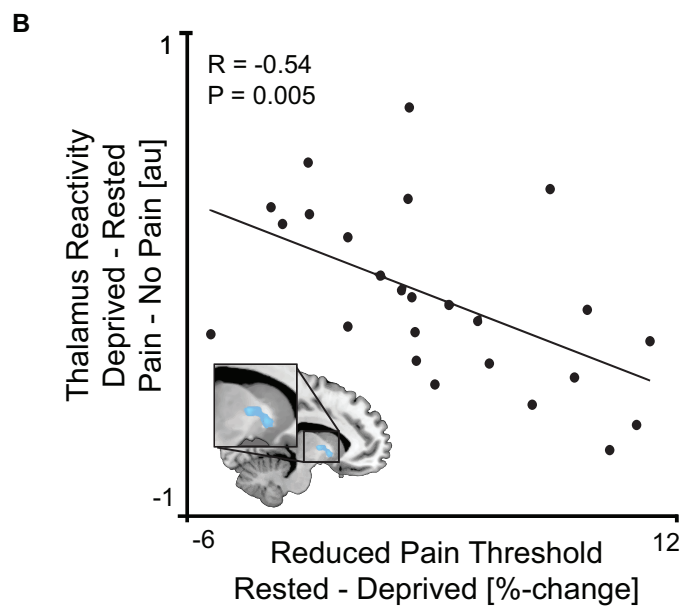
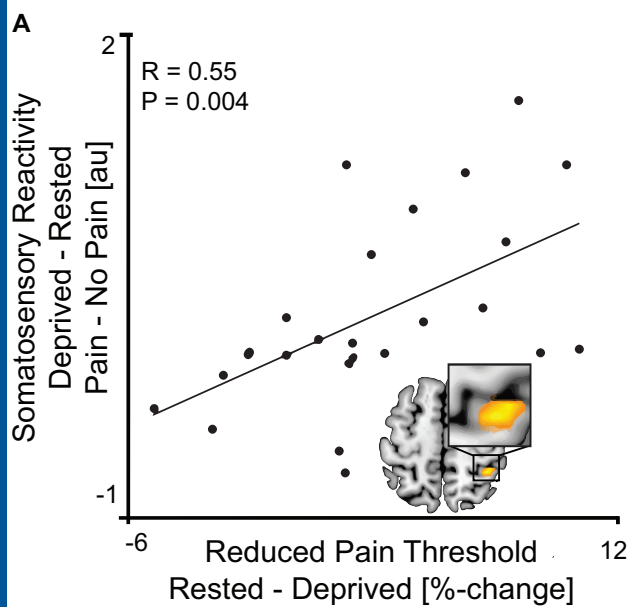


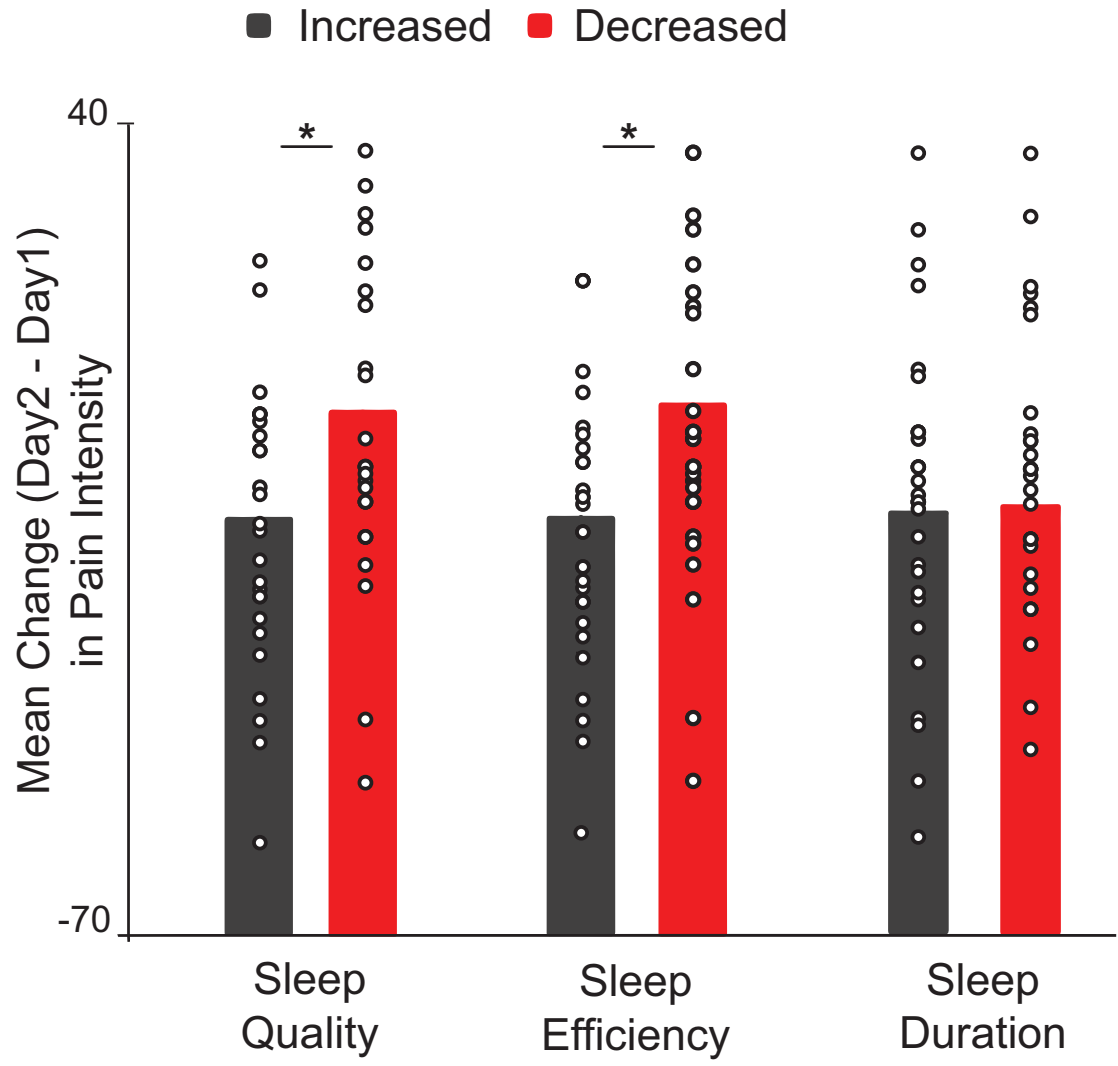
Fig. 5. Online study results. Bar graph representing participants experiencing a decrease (red bar) or increase (black) in measures of sleep efficiency (SE) (Night 2 – Night 1 mean decreased SE = -3.8% [$\pm 5.6\%$ s.d.]; Night 2 – Night 1 mean increased SE = 10.1% [$\pm 11.1\%$ s.d.]), quality (SQ) (Night 2 – Night 1 mean decreased SQ = -0.48 [± 0.75 s.d.]; Night 2 – Night 1 mean increased SQ = 1.7 [± 0.81 s.d.]), and duration (Night 2 – Night 1 mean decreased duration = -54.375 minutes [± 45.7 s.d.]; Night 2 – Night 1 mean increased duration = 90.7 minutes [± 48.9 s.d.]). Each point represents a single-subject's change in reported pain from day 1 to day 2. Night-to-night decreases in sleep efficiency and quantity, but not quantity, resulted in corresponding higher pain from one day to the next, relative to those with night-to-night increases in these sleep measures. * $P < 0.05$.











Sleep Deprivation > Sleep Rested						
Region (L/R)	Radius(mm)	X	Y	Z	P	R
Somatosensory (R)	8	36	-45	59	0.004	0.55
Dorsal ACC (R)	8	16	-7	39	0.02	0.04
Superior Parietal (R)	5	26	-42	66	0.51	-0.04
Sleep Rested > Sleep Deprived						
Region (L/R)	Radius(mm)	X	Y	Z	P	R
Middle Insula (R)	8	32	4	11	0.0005	-0.02
Thalamus (L)	4	-10	6	10	0.005	-0.54
Anterior Insula (L)	8	-27	25	0	0.005	0.24
Nucleus Accumbens	6	±9	2	-7	0.005	0.23
Operculum (R)	5	57	1	5	0.04	-0.36
Substantia Nigra (R)	4	10	-8	12	0.09	-0.31
Caudate	4	±18	-2	24	0.38	-0.32
IFG (R)	4	64	24	-6	0.47	-0.47
Middle Cingulate / SMA	10	6	-6	48	0.48	0.01
Precuneus (R)	4	4	-54	48	0.60	0.32
ACC (R)	6	12	28	28	0.63	0.36
DLPFC (R)	8	33	35	42	0.90	0.10

Table 1. Region of interest analysis. Brain ROI analysis showing center (MNI coordinates) and radius (millimeters) of extracted spheres. Bolded rows are significant, following FDR correction, for paired comparison (sleep rested </> sleep deprived) of pain-related brain activity. Column R shows Pearson correlations between pain-related brain changes (sleep deprived – sleep rested) and changes in thermal pain threshold (sleep rested – sleep deprived), and bolded rows are significant following FDR correction. ACC, anterior cingulate cortex; IFG, inferior frontal gyrus; SMA, supplementary motor area; DLPFC, dorsolateral prefrontal cortex.